Mathematics Bootcamp — Proofs

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Blackboard companion to the bootcamp
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The slides carry the statements; these are the arguments behind them. Everything here is meant for the board.

1. Day 1 — Choice and Rationalization

Pairs recover revealed preference

Supports: "WARP and Rationalizability".

Suppose every pair belongs to $\mathbb{B}$ and choice is nonempty. Under WARP,

$$x \succeq^* y \quad \Longleftrightarrow \quad x \in C(\{x, y\}).$$

Proof. ($\Leftarrow$) The pair $\{x, y\}$ is itself an observed menu containing both alternatives, and x is chosen from it. Hence $x \succeq^* y$.
($\Rightarrow$) Let $x \succeq^* y$, so $x \in C(B)$ for some B with $x, y \in B$. Since $C(\{x, y\}) \neq \varnothing$, either $x \in C(\{x, y\})$ — and we are done — or $y \in C(\{x, y\})$. In the latter case WARP, applied to B and $\{x, y\}$, forces $x \in C(\{x, y\})$ as well.
Completeness. Every pair menu chooses something, so for any x, y at least one of $x \in C(\{x, y\})$, $y \in C(\{x, y\})$ holds; hence $x \succeq^* y$ or $y \succeq^* x$. ■

Triples rule out cycles

Supports: "WARP and Rationalizability".

Suppose the pair menus give

$$C(\{x, y\}) = \{x\}, \qquad C(\{y, z\}) = \{y\}, \qquad C(\{x, z\}) = \{z\}.$$

No two of these menus contain the same pair, so WARP never applies across them: with pairs alone the cycle x over y, y over z, z over x is perfectly consistent with the axiom.
Proof. Add the triple $T = \{x, y, z\}$, which a rich domain supplies. Since $C(T) \neq \varnothing$, T chooses at least one alternative.

- If $x \in C(T)$: both x and z lie in T and in $\{x, z\}$, and $z \in C(\{x, z\})$. WARP forces $x \in C(\{x, z\})$, contradicting $C(\{x, z\}) = \{z\}$.
- If $y \in C(T)$: the same argument against $C(\{x, y\}) = \{x\}$.
- If $z \in C(T)$: the same argument against $C(\{y, z\}) = \{y\}$.

Every case fails, so $C(T) = \varnothing$ — a contradiction. The cycle is therefore impossible, and $\succeq^*$ is transitive. ■

WARP is equivalent to rationalizability

Supports: "WARP and Rationalizability".

Theorem. Suppose $\mathbb{B}$ contains every nonempty $B \subseteq X$ with $|B| \le 3$, and $C(B) \neq \varnothing$ for every $B \in \mathbb{B}$. Then C is rationalizable if and only if it satisfies WARP.

Proof. *Necessity.* Let a rational $\succeq$ rationalize C. Take $x, y \in B$ with $x \in C(B)$, and B' with $x, y \in B'$ and $y \in C(B')$. From B we get $x \succeq y$; from B', $y \succeq z$ for every $z \in B'$. Transitivity gives $x \succeq z$ for every $z \in B'$, so $x \in C(B')$. That is WARP.

Sufficiency. Let $\succeq^*$ be the revealed preference relation. By the first lemma it is complete and satisfies $x \succeq^* y \iff x \in C(\{x, y\})$; by the second it is transitive. So $\succeq^*$ is rational. It remains to check $C(B) = \{x \in B : x \succeq^* y$ for every $y \in B\}$.

- $\subseteq$ If $x \in C(B)$ then $x \succeq^* y$ for every $y \in B$, directly from the definition of $\succeq^*$.
- $\supseteq$ Let $x \succeq^* y$ for every $y \in B$, and pick any $y \in C(B)$, which is nonempty. Then $x \succeq^* y$ means the pair menu $\{x, y\}$ chooses x. WARP, applied to $\{x, y\}$ and B with $y \in C(B)$, forces $x \in C(B)$.

■

The key step is the last one: if $y \in C(B)$ and $x \succeq^* y$, the pair menu chooses x, and WARP carries that back into B.

2. Day 3 — Properties of Preferences

Monotonicity implies local nonsatiation

Supports: "Monotonicity and Local Nonsatiation".

Proposition. On $X = \mathbb{R}^L_+$, monotonicity implies local nonsatiation.

Proof. Let $x \in \mathbb{R}^L_+$ and $\varepsilon > 0$ be arbitrary. Put $y = x + \frac{\varepsilon}{L}\mathbf{1}$. Then $y \in \mathbb{R}^L_+$ and $y \gg x$, so monotonicity gives $y \succ x$. Moreover

$$\|x - y\| = \sqrt{\sum_{i=1}^{L} \left(\tfrac{\varepsilon}{L}\right)^2} = \sqrt{L\left(\tfrac{\varepsilon}{L}\right)^2} = \frac{\varepsilon}{\sqrt{L}} \le \varepsilon.$$

So every ε-neighborhood of x contains a strictly better bundle. ■

The converse fails: local nonsatiation does not imply monotonicity.

Convex preferences and quasiconcave utility

Supports: "Convex Preferences and Quasiconcave Utility" (Kreps 2.8b).

Let X be convex and let u represent $\succeq$.

Proof. For each $x \in X$,

$$U(x) = \{y \in X : y \succeq x\} = \{y \in X : u(y) \ge u(x)\},$$

because u represents $\succeq$. So the upper contour set of $\succeq$ at x *is* the upper level set of u at height $u(x)$. Now $\succeq$ is convex exactly when every $U(x)$ is convex, and u is quasiconcave exactly when every upper level set is convex — and these are the same family of sets. Replacing the weak inequalities by strict ones throughout gives the strict version. ■

Lexicographic preferences have no representation

Supports: "Lexicographic Preferences Have No Representation" (MWG 3.C.1).

Let $X = \mathbb{R}^2_+$ and $x \succeq y$ iff $x_1 > y_1$, or $x_1 = y_1$ and $x_2 \ge y_2$.

Proof. Suppose some u represents $\succeq$. For each x_1 we have $(x_1, 2) \succ (x_1, 1)$, hence $u(x_1, 2) > u(x_1, 1)$. By the Density Theorem (Bartle 2.4.8) choose a rational $r(x_1)$ with

$$u(x_1, 2) > r(x_1) > u(x_1, 1).$$

The map r is injective: if $x_1 > x_1'$ then $(x_1, 1) \succ (x_1', 2)$, so

$$r(x_1) > u(x_1, 1) > u(x_1', 2) > r(x_1').$$

So r is a one-to-one map from $\mathbb{R}$ into $\mathbb{Q}$. But $\mathbb{R}$ is uncountable and $\mathbb{Q}$ is countable (Bartle 1.3.6) — impossible. ■

This is why continuity is needed: lexicographic preferences are complete, transitive, strongly monotone and strictly convex, and still admit no representation at all.

A continuous representation gives continuous preferences

Supports: "A Continuous Representation Gives Closed Contours".

Proof. If $f : X \to \mathbb{R}$ is continuous then $f^{-1}([c, \infty))$ and $f^{-1}((-\infty, c])$ are closed relative to X for every $c \in \mathbb{R}$. Apply this to a continuous representation u of $\succeq$:

$$U(x) = u^{-1}\big([u(x), \infty)\big), \qquad L(x) = u^{-1}\big((-\infty, u(x)]\big)$$

are then closed for every x. By the contour-set characterization (Kreps 1.14), $\succeq$ is continuous. ■

3. Day 5 — General Equilibrium

Walras' law

Supports: "Market Clearing".

Theorem. Suppose that at prices $\{w_t, r_t\}$ the allocations $\{c_t, a_{t+1}\}$ and $\{k_t, n_t\}$ solve the household and the firm problem, and that in every period the labor and asset markets clear, $n_t = 1$ and $a_t = k_t$. Then the goods market clears as well, in every period.

Proof. Because the allocation solves the household problem it satisfies the budget constraint,

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t.$$

Asset-market clearing gives $a_t = k_t$ and $a_{t+1} = k_{t+1}$, so

$$c_t + k_{t+1} = w_t + (1 + r_t)k_t, \qquad \text{that is} \qquad c_t + k_{t+1} - (1 - \delta)k_t = w_t + (r_t + \delta)k_t.$$

The firm's first-order conditions set each factor price equal to its marginal product. With constant returns to scale, Euler's theorem for homogeneous functions gives

$$w_t n_t + (r_t + \delta)k_t = A_t f(k_t, n_t),$$

so competitive profits are zero. Using $n_t = 1$,

$$c_t + k_{t+1} - (1 - \delta)k_t = w_t n_t + (r_t + \delta)k_t = A_t f(k_t, n_t),$$

which is exactly goods-market clearing, $c_t + k_{t+1} = A_t f(k_t, n_t) + (1 - \delta)k_t$. ■

Counting: with $T + 1$ periods there are $3(T + 1)$ clearing conditions but only $2(T + 1)$ prices. Walras' law resolves the apparent mismatch — one condition per period is redundant.